ASSIGNMENT-1

Foundations of Cryptography

Q1. Define Cryptography and Its Three Main Goals

Cryptography is the technique of protecting information by changing readable data into a protected form so unauthorized people cannot understand or change it. It uses mathematical methods, keys, and algorithms to secure information during storage or communication.

Three main goals:
• **Confidentiality:** Only authorized people should read the information.
• **Integrity:** Information should not be changed without permission.
• **Authentication:** The identity of a sender or receiver should be verified.

Cryptography is therefore an important part of computer and network security. It protects information from unauthorized access and helps users communicate safely over digital networks.

Q2. Plaintext and Ciphertext

Plaintext is the original readable information before encryption. **Ciphertext** is the unreadable form produced after encryption using an algorithm and a key.

Example:
Plaintext: **HELLO**
After encryption: **KHOOR**

Here, HELLO is plaintext and KHOOR is ciphertext. The receiver uses the correct key and decryption process to convert ciphertext back into the original plaintext. In simple words, plaintext is the message we can understand, while ciphertext is the protected version.

Q3. Cryptography in the Modern Digital World

Cryptography is a method of protecting digital information using mathematical techniques. It converts readable information into a protected form and allows authorized users to recover or verify it. Cryptography is essential today because personal, financial, and business information is constantly transferred through the internet.

Importance:
• Protects confidential information from unauthorized access.
• Helps detect unauthorized changes to data.
• Helps verify users and systems.

Real-life applications:
1. **Banking:** Online banking and digital payments use cryptographic techniques to protect financial information.
2. **Messaging Apps:** Secure messaging can use encryption so unauthorized people cannot easily read messages.
3. **E-commerce:** Cryptography helps protect payment and account information during online shopping.

Thus, cryptography is a basic requirement for safe digital communication.

Q4. Difference Between Encryption and Decryption

Encryption converts readable data, called plaintext, into an unreadable form, called ciphertext, using an algorithm and a key. Its purpose is to protect information from unauthorized access.

Decryption is the reverse process. It converts ciphertext back into readable plaintext using the appropriate key and decryption algorithm.

Simple process:
Plaintext → Encryption → Ciphertext
Ciphertext → Decryption → Plaintext

Therefore, encryption protects the message while it is stored or transmitted, and decryption allows an authorized receiver to read the protected message.

Q5. Symmetric and Asymmetric Key Cryptography

There are two major types of cryptography: symmetric key and asymmetric key cryptography.

Symmetric Key Cryptography
• Uses the same secret key for encryption and decryption.
• Generally fast and suitable for large amounts of data.
• The secret key must be shared securely.
• Examples: **AES and DES**.

Asymmetric Key Cryptography
• Uses two related keys: a public key and a private key.
• Generally slower than symmetric encryption.
• The public key can be shared, while the private key remains secret.
• Examples: **RSA and ECC**.

Comparison: Symmetric cryptography is mainly used for fast data encryption, while asymmetric cryptography is commonly used for authentication, digital signatures, and secure key exchange. Modern systems often combine both methods.

Q6. Hashing and Encryption

Hashing converts data into a fixed-length value called a **hash**. A good cryptographic hash function is designed to be one-way, so recovering the original data from the hash should be difficult.

Encryption is different because it is designed to be **reversible** when the correct key is available. Encrypted data can be decrypted to recover the original message.

Hashing: Data → Hash
Encryption: Plaintext → Ciphertext → Decryption → Plaintext

Hashing is commonly used for checking data integrity and password verification. Encryption is mainly used when information needs to remain confidential and later be recovered by an authorized user.

Q7. Why Confidentiality, Integrity and Non-Repudiation Are Challenging Together

Achieving confidentiality, integrity, and non-repudiation together is challenging because each goal requires different security mechanisms. **Confidentiality** requires encryption, while **integrity** requires cryptographic hashes or authenticated encryption. **Non-repudiation** generally requires digital signatures and secure private-key management.

In a real communication system, keys must be generated, stored, distributed, and protected correctly. If a private key is stolen, a false signature may be created. If encryption is used incorrectly, confidential information may still be exposed. Systems also need trusted identities and secure certificate management. Therefore, achieving all three goals requires secure protocols, correct implementation, key management, and protection against attacks.

Q8. Role of Cryptography in Network Security

Cryptography protects information while it travels between computers and users. It helps provide confidentiality, integrity, authentication, and non-repudiation.

1. Confidentiality: Encryption prevents unauthorized people from reading network data. **Example:** HTTPS protects information exchanged between a browser and a website.

2. Integrity: Hashes and authentication mechanisms help detect whether data has been changed. **Example:** A secure protocol can verify that transmitted data was not modified.

3. Authentication: Digital certificates and signatures help verify identities. **Example:** A browser can use a website certificate to authenticate the website.

4. Non-repudiation: Digital signatures can provide evidence that a key holder signed data. **Example:** A digitally signed electronic document can be verified later.

Thus, cryptography is a foundation of secure network communication.

Q9. Two Network Security Protocols

Two network security protocols that use cryptographic techniques are **SSL/TLS** and **IPSec**.

1. SSL/TLS: TLS protects communication between applications over a network. It is commonly used with HTTPS to protect communication between a web browser and a web server. It can provide encryption, integrity protection, and authentication using certificates and cryptographic algorithms.

2. IPSec: IPSec is a collection of protocols used to secure IP network communication. It can provide confidentiality, integrity, authentication, and protection against certain network attacks. It is commonly used in secure network connections such as VPNs.

Both use cryptographic techniques to make network communication more secure.

Q10. Evolution of Cryptography and Impact of Computing

Cryptography evolved from simple classical ciphers to complex computer-based security systems. Early methods such as the **Caesar Cipher** used simple letter substitutions and were easy to break. The

Vigenère Cipher improved this idea by using multiple substitutions, but it could still be attacked using statistical methods.

During World War II, the **Enigma machine** showed the importance and complexity of machine-based encryption. Later, **DES** provided standardized computer-based encryption, but increasing computing power eventually made its 56-bit key too weak.

Modern algorithms such as **AES** use stronger designs and larger keys. Public-key systems such as **RSA** also introduced new methods for key management and digital signatures.

As computing power increases, attacks become faster, so cryptographic algorithms must improve. Quantum computing also creates future challenges for systems such as RSA.

Q11. Historical Development of Cryptographic Techniques

The history of cryptography began with classical methods. The **Caesar Cipher** shifted letters by a fixed number of positions, while the **Vigenère Cipher** used a keyword to create different substitutions. These methods were useful historically but were not strong enough against advanced analysis.

During **World War II**, the German **Enigma machine** used rotating components to create changing substitutions. Its analysis and eventual breaking became an important milestone in cryptography and computing.

With modern computers, stronger algorithms were developed. **DES** became an important encryption standard, but its 56-bit key eventually became inadequate against increasing computing power. **AES** was later selected as a stronger replacement.

Another major turning point was **public-key cryptography**. Diffie and Hellman's 1976 work introduced the public-key concept and key exchange, and **RSA** was developed in 1977. Modern cryptography combines strong mathematical algorithms, secure protocols, and careful key management.

Q12. Introduction of Public-Key Cryptography

The concept of public-key cryptography was publicly introduced by **Whitfield Diffie and Martin Hellman in 1976**. Their work introduced the idea of using related public and private keys and described the Diffie-Hellman key exchange method.

The main purpose was to solve the **key distribution problem** found in traditional symmetric cryptography. In a symmetric system, two parties need to share a secret key before secure communication can begin. Public-key cryptography made it possible to use a public key that could be openly distributed while keeping a private key secret.

This development became a major turning point in modern cryptography and later supported technologies such as digital signatures, certificates, secure key exchange, and secure internet communication.